The Fundamental Theorem of Calculus

Section 1: FTC Part 1 (Derivative of an Accumulation Function)

1 If $g(x) = \int_1^x \sqrt{t^3 + 1}\, dt$, find $g'(x)$.

a. $\sqrt{x^3 + 1}$

b. $\frac{3x^2}{2\sqrt{x^3 + 1}}$

c. $\sqrt{x^3 + 1} - \sqrt{2}$

d. $\frac{1}{3}(x^3 + 1)^{3/2}$

2 Find $\frac{d}{dx}\left[\int_x^5 \cos(t^2)\, dt\right]$.

a. $-\cos(x^2)$

b. $\cos(x^2)$

c. $-2x\sin(x^2)$

d. $\cos(25) - \cos(x^2)$

3 Find $h'(x)$ for $h(x) = \int_0^{x^2} \sin(t)\, dt$.

a. $2x\sin(x^2)$

b. $\sin(x^2)$

c. $2x\cos(x^2)$

d. $\sin(x^4)$

Section 2: FTC Part 1 with Variable Limits on Both Bounds

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Find $F'(x)$ for $F(x) = \int_{2x}^{3x} (t^2 + 1)\,dt$.

a. $27x^2 + 3 - 2(4x^2 + 1) = 19x^2 + 1$

b. $19x^2 + 1$

c. $27x^2 - 8x^2 = 19x^2$

d. $5x^2 + 1$

5 Evaluate $\frac{d}{dx}\left[\int_{\sin x}^{\cos x} e^t\,dt\right]$.

a. $-e^{\cos x}\sin x - e^{\sin x}\cos x$

b. $e^{\cos x} - e^{\sin x}$

c. $e^{\cos x}\sin x + e^{\sin x}\cos x$

d. $-e^{\cos x} + e^{\sin x}$

6 Find the inflection points of $g(x) = \int_0^x (t^3 - 3t)\,dt$.

a. $x = 1$ and $x = -1$

b. $x = 0$ and $x = \pm\sqrt{3}$

c. $x = 3$

d. No inflection points

Section 3: FTC Part 2 (Evaluating Definite Integrals)

7 Evaluate $\int_1^3 (3x^2 - 4x + 1)\,dx$.

a. 12

b. 14

c. 10

d. 16

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Evaluate $\int_0^{\pi/4} \sec^2(x)\,dx$.

a. 1

b. $\sqrt{2}$

c. 1/2

d. 0

9 Evaluate $\int_1^e \frac{4}{x}\,dx$.

a. 4

b. $4e$

c. $4\ln(e) - 4 = 0$

d. 1

Section 4: Average Value of a Function

10 State the formula for the average value f_{avg} of a continuous function f on $[a, b]$.

a. $f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx$

b. $f_{\text{avg}} = \frac{f(a)+f(b)}{2}$

c. $f_{\text{avg}} = \int_a^b f(x)\,dx$

d. $f_{\text{avg}} = \frac{f(b)-f(a)}{b-a}$

11 Find the average value of $f(x) = 3x^2 - 2x$ on the interval $[0, 2]$.

a. 2

b. 4

c. 3

d. 1

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 By the Mean Value Theorem for Integrals, find $c \in [0, 3]$ such that $f(c) = f_{\text{avg}}$ for $f(x) = 2x$.

a. 1.5

b. 1.0

c. 2.0

d. 0.75

© ClipSAT: By Mr. Mohamed Abdallah +966597688647